AFRICAST

TIDY TIME SERIES & FORECASTING IN R

ARIMA models

Session 6 of 10



SECTION 1 OF 3

ARIMA models

1 **ARIMA models** 2 Seasonal ARIMA models 3 Forecast ensembles



ARIMA models

ARIMA models describe the autocorrelations in the data: how each observation relates to past observations and past errors.

An ARIMA model is rarely interpretable in terms of visible data structures like trend and seasonality. But it can capture a huge range of time series patterns.

Stationarity

Definition

If $\{y_t\}$ is a stationary time series, then for all s, the distribution of $(y_t, \ldots, y_{t+s})$ does not depend on t.

A **stationary series** is:

- roughly horizontal
- constant variance
- no patterns predictable in the long-term

Stationary?



Yes **0** 0%

No **0** 0%

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0 responses

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Stationary?



Yes **0** 0%

No **0** 0%

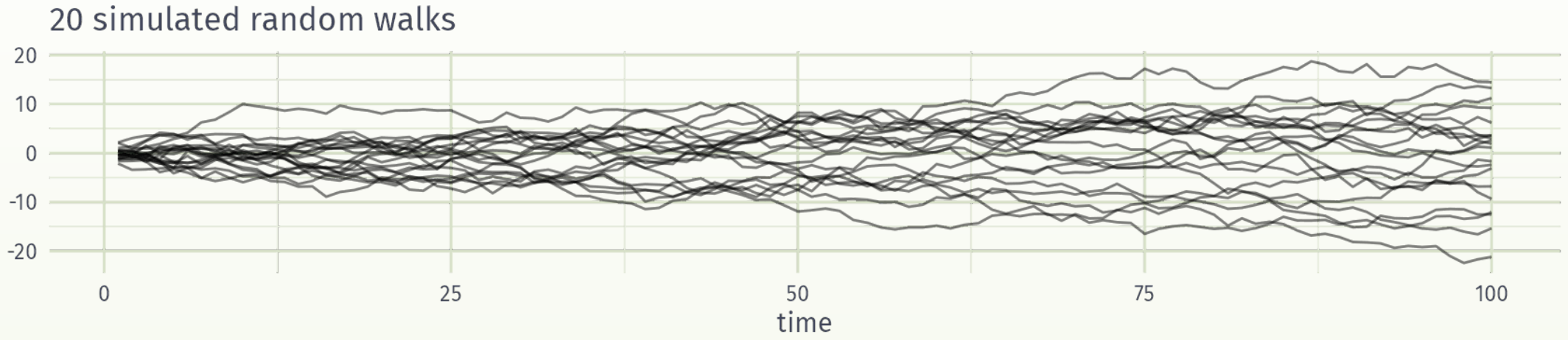
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Random walks

A random walk, $y_t = y_{t-1} + \varepsilon_t$ with white noise ε_t, has no trend or seasonality.



Yes **0** 0%

No **0** 0%

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0 responses

Differencing

Differences

First difference: $y'_t = y_t - y_{t-1}$

Second difference: $y''_t = y'_t - y'_{t-1} = y_t - 2y_{t-1} + y_{t-2}$

Seasonal difference: $y_t - y_{t-m}$, where m is the seasonal period

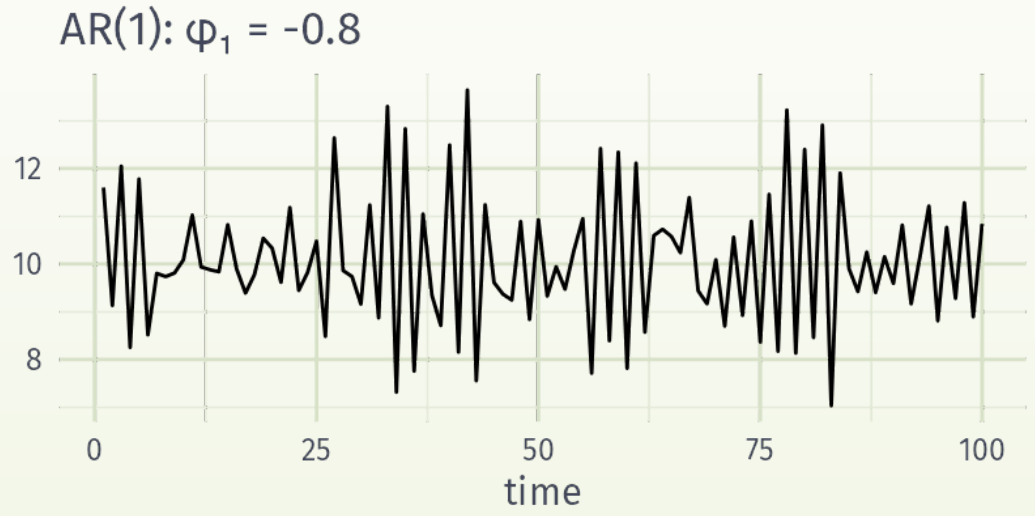
- Differencing helps to **stabilise the mean**.
- A random walk is not stationary (its variance grows over time), but its first difference is white noise, which is stationary. The Google stock price behaves like a random walk.
- Occasionally a second difference is needed; it is almost never necessary to go further.

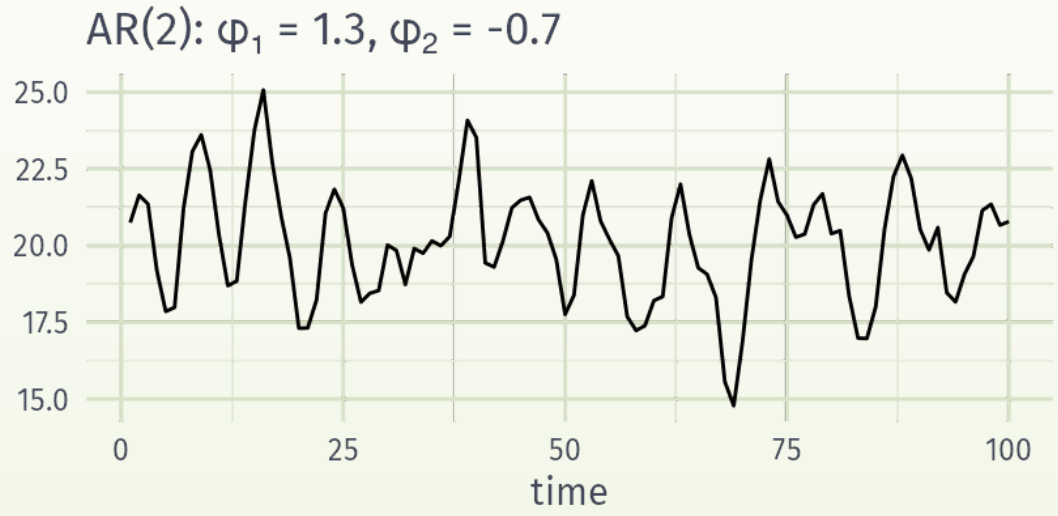
Autoregressive models

Autoregressive (AR) models

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t,$$

where ε_t is white noise. A multiple regression with **lagged values** of y_t as predictors.





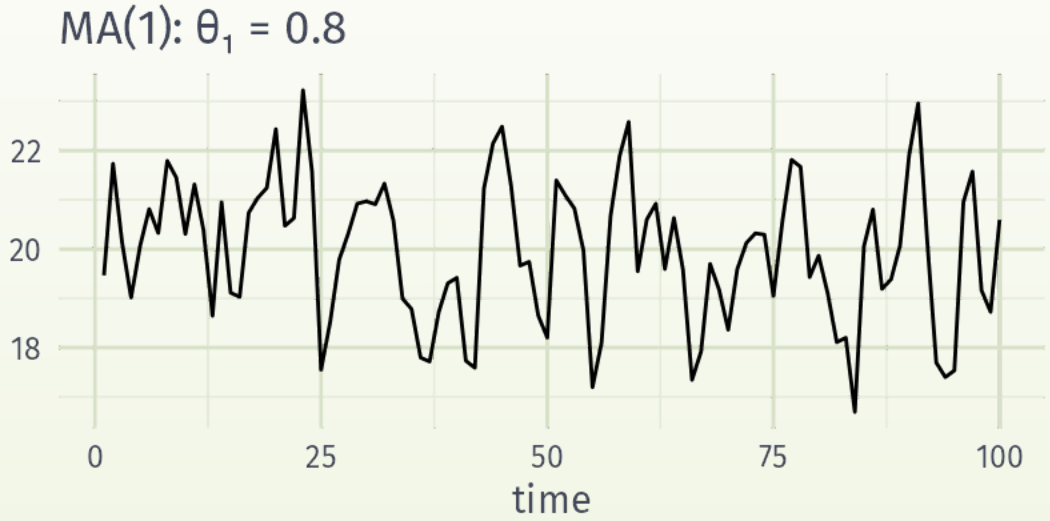
- Cyclic behaviour is possible when $p \geq 2$.

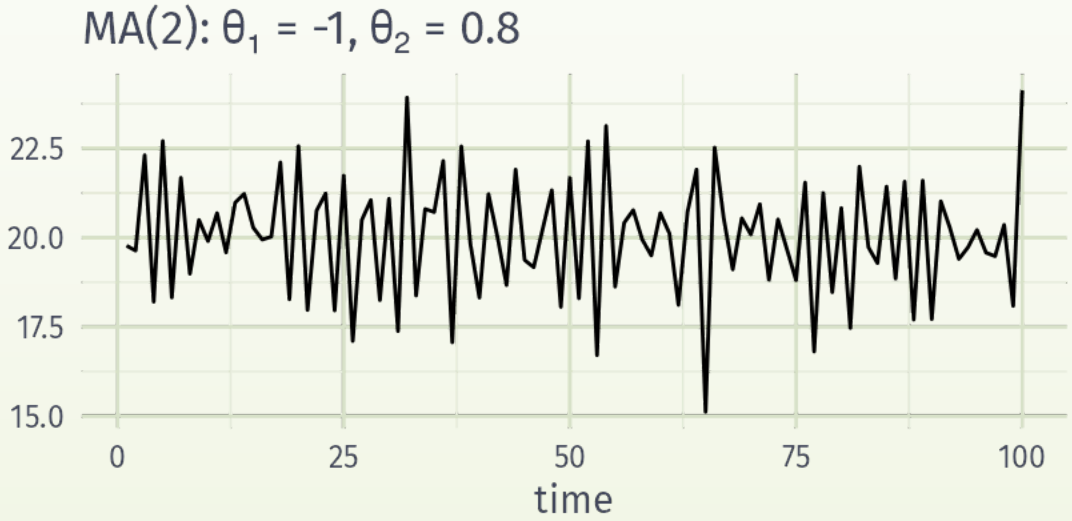
Moving average models

Moving average (MA) models

$$y_t = c + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q},$$

where ε_t is white noise. A multiple regression with **lagged errors** as predictors. Don't confuse with moving average smoothing!





ARIMA models

Autoregressive moving average (ARMA) models

$$y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q} + \varepsilon_t.$$

- Predictors include both lagged values of y_t and lagged errors.

Autoregressive integrated moving average models

- Combine ARMA model with differencing.
- d-differenced series follows an ARMA model.
- Need to choose p, d, q and whether or not to include c.

ARIMA models

ARIMA(p, d, q) model

AR:	$p =$ order of the autoregressive part
I:	$d =$ degree of first differencing involved
MA:	$q =$ order of the moving average part.

- White noise model: ARIMA(0,0,0)
- Random walk: ARIMA(0,1,0) with no constant
- Random walk with drift: ARIMA(0,1,0) with const.
- AR(p): ARIMA(p,0,0)
- MA(q): ARIMA(0,0,q)

Example: National populations

```
fit_pop <- global_economy |>
  model(arima = ARIMA(Population))
fit_pop
```

```
# A mable: 263 x 2
# Key:     Country [263]
   Country                          arima
   <fct>                          <model>
 1 Afghanistan                <ARIMA(4,2,1)>
 2 Albania                    <ARIMA(0,2,2)>
 3 Algeria                    <ARIMA(2,2,2)>
 4 American Samoa             <ARIMA(2,2,2)>
 5 Andorra             <ARIMA(2,1,2) w/ drift>
 6 Angola                     <ARIMA(4,2,1)>
 7 Antigua and Barbuda <ARIMA(2,1,2) w/ drift>
 8 Arab World                 <ARIMA(0,2,1)>
 9 Argentina                  <ARIMA(2,2,2)>
10 Armenia                    <ARIMA(3,2,0)>
# i 253 more rows
```

Example: National populations

```
fit_pop |>
  filter(Country == "Australia") |>
  report()
```

```
Series: Population
Model: ARIMA(0,2,1)

Coefficients:
          ma1
      -0.661
s.e.   0.107

sigma^2 estimated as 4.063e+09:  log likelihood=-699
AIC=1401   AICc=1402   BIC=1405
```

With $d = 2$ the MA(1) model is for the second differences $y_t - 2y_{t-1} + y_{t-2}$, so

$$y_t = 2y_{t-1} - y_{t-2} - 0.661\varepsilon_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim \text{NID}(0, 4.06 \times 10^9)$$

Example: National populations

Australia’s population model is ARIMA(0,2,1) with no constant. What shape will its long-term forecasts take?

Flat line	0	0%
Straight line	0	0%
Quadratic curve	0	0%
Decay to zero	0	0%

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0 responses

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Understanding ARIMA models

c is the constant in the ARMA model for the d-times differenced series.

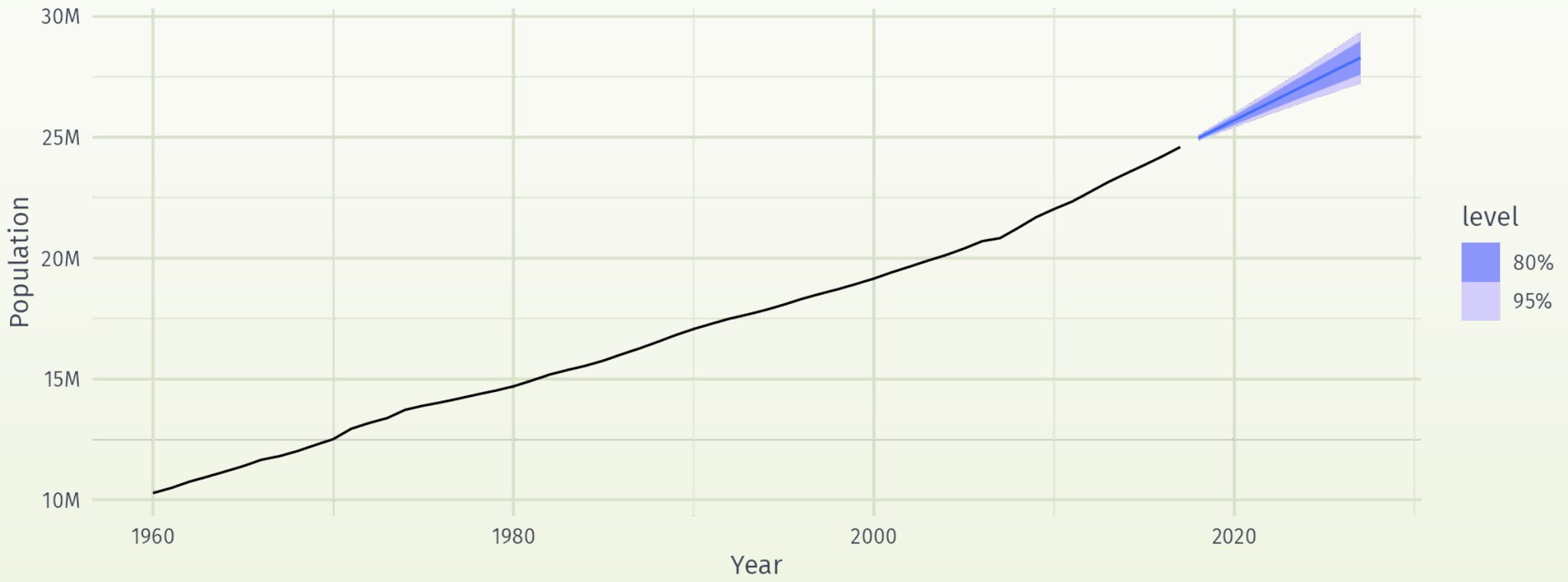
- If $c = 0$ and $d = 0$, the long-term forecasts will go to zero.
- If $c = 0$ and $d = 1$, the long-term forecasts will go to a non-zero constant.
- If $c = 0$ and $d = 2$, the long-term forecasts will follow a straight line.
- If $c \neq 0$ and $d = 0$, the long-term forecasts will go to the mean of the data.
- If $c \neq 0$ and $d = 1$, the long-term forecasts will follow a straight line.
- If $c \neq 0$ and $d = 2$, the long-term forecasts will follow a quadratic trend.

Forecast variance and d

- The higher the value of d, the more rapidly the prediction intervals increase in size.
- For $d = 0$, the long-term forecast standard deviation will go to the standard deviation of the historical data.

Example: National populations

```
fit_pop |>
  forecast(h = 10) |>
  filter(Country == "Australia") |>
  autoplot(global_economy) +
  scale_y_continuous(labels = scales::label_number(scale = 1e-6, suffix = "M"))
```



How does ARIMA() work?

Hyndman and Khandakar (JSS, 2008) algorithm

- Select the number of seasonal differences D via a seasonal strength test, then the number of differences d via KPSS tests.
- Select p, q (and P, Q for seasonal models) and inclusion of c by minimising AICc.
- Use stepwise search to traverse model space.

How does ARIMA() work?

$$\text{AICc} = -2\log(L) + 2(p+q+k+1)\left[1 + \frac{(p+q+k+2)}{T-p-q-k-2}\right]$$

where L is the maximised likelihood fitted to the *differenced* data, $k = 1$ if $c \neq 0$ and $k = 0$ otherwise.

Yes **0** 0%

No **0** 0%

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0 responses

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How does ARIMA() work?

1. Fit ARIMA$(2, d, 2)$, ARIMA$(0, d, 0)$, ARIMA$(1, d, 0)$ and ARIMA$(0, d, 1)$. The lowest AICc becomes the current model.
2. Fit variations of the current model: vary p, q or both by ± 1, and include/exclude c. The lowest AICc becomes the current model.
3. **Repeat step 2 until no lower AICc can be found.**

Yes **0** 0%

No **0** 0%

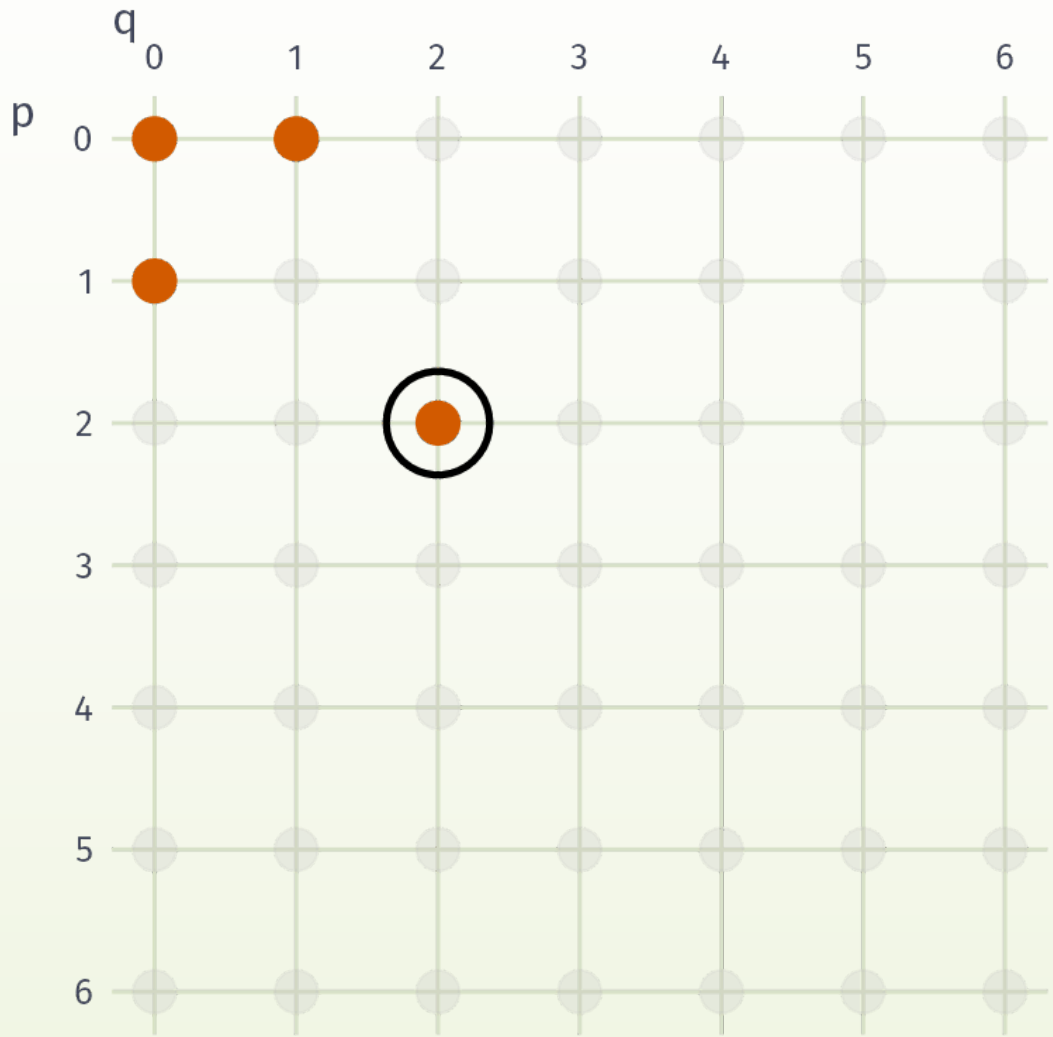


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0 responses

● CLOSED 0 connected

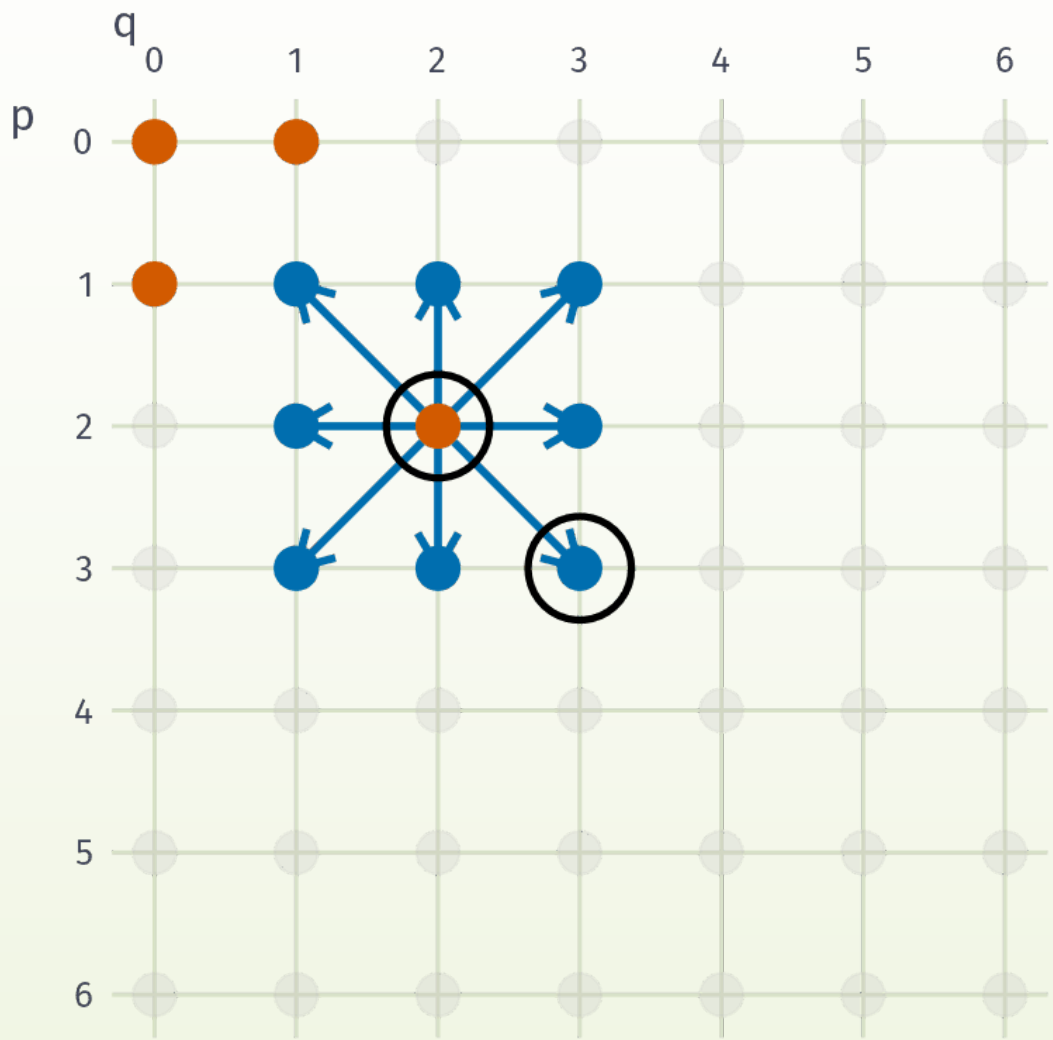
How does ARIMA() work?



Step 1: fit the four starting models.

ARIMA$(2, d, 2)$ has the lowest AICc, so it becomes the current model (circled).

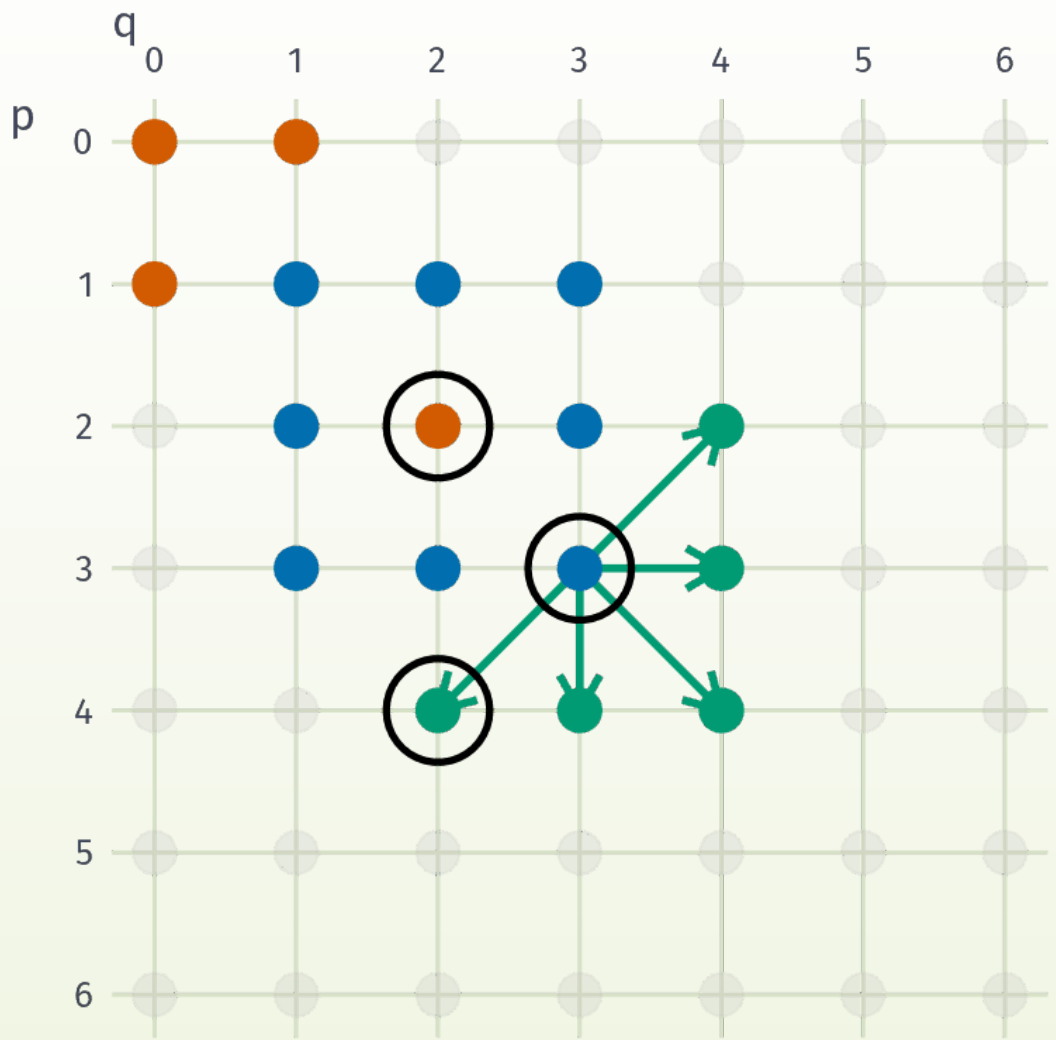
How does ARIMA() work?



Step 2: fit the neighbours of ARIMA $(2, d, 2)$.

ARIMA$(3, d, 3)$ has a lower AICc, so it becomes the current model.

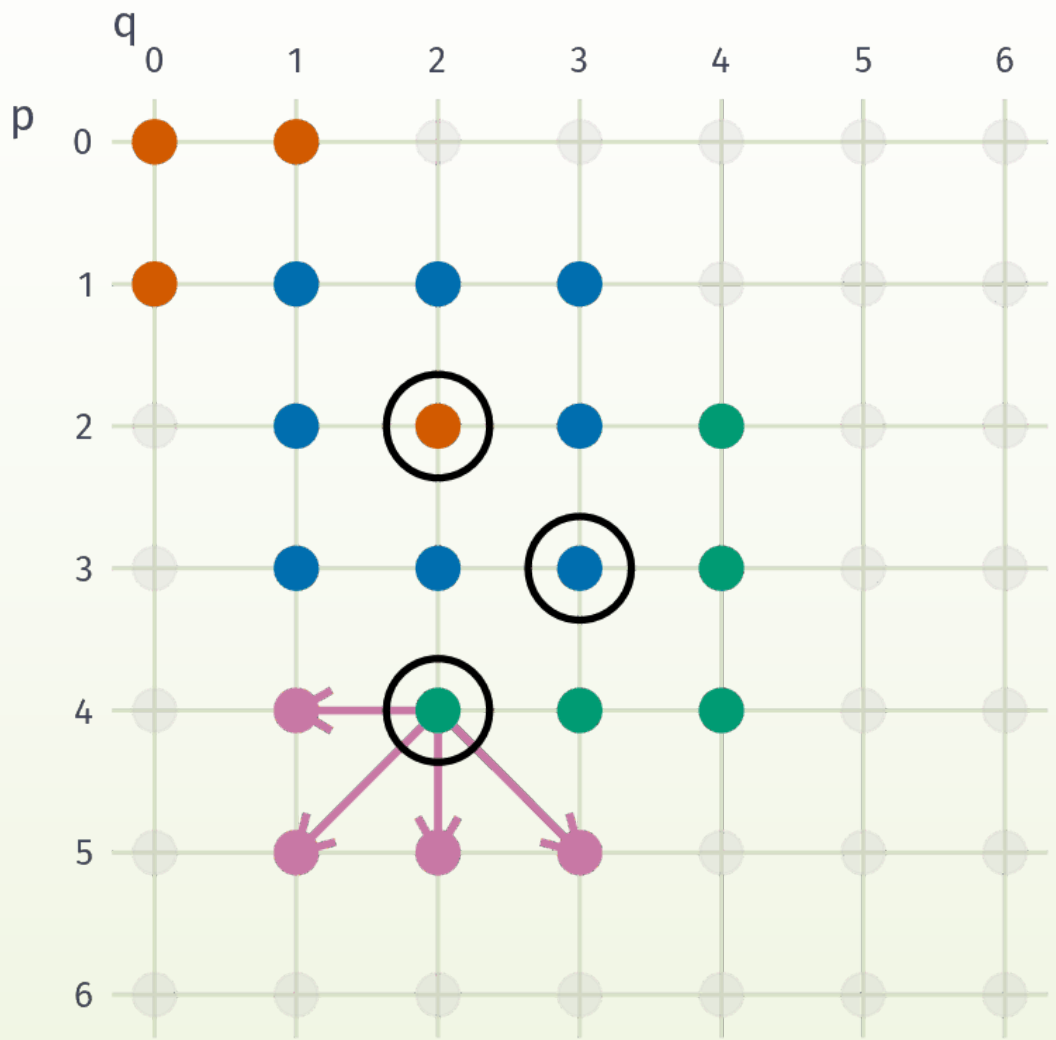
How does ARIMA() work?



Step 2 (repeated): fit the neighbours of ARIMA$(3, d, 3)$.

ARIMA$(4, d, 2)$ has a lower AICc, so it becomes the current model.

How does ARIMA() work?



Step 2 (repeated): fit the neighbours of ARIMA$(4, d, 2)$.

None has a lower AICc, so the search stops and ARIMA$(4, d, 2)$ is selected.

The search is local, so a model with a lower AICc may exist elsewhere in the grid.

SECTION 2 OF 3

Seasonal ARIMA models

1 ARIMA models
2 **Seasonal ARIMA models**
3 Forecast ensembles



Seasonal ARIMA models

$$\text{ARIMA} \quad \underbrace{(p, d, q)}_{\text{Non-seasonal part}} \quad \underbrace{(P, D, Q)_m}_{\text{Seasonal part}}$$

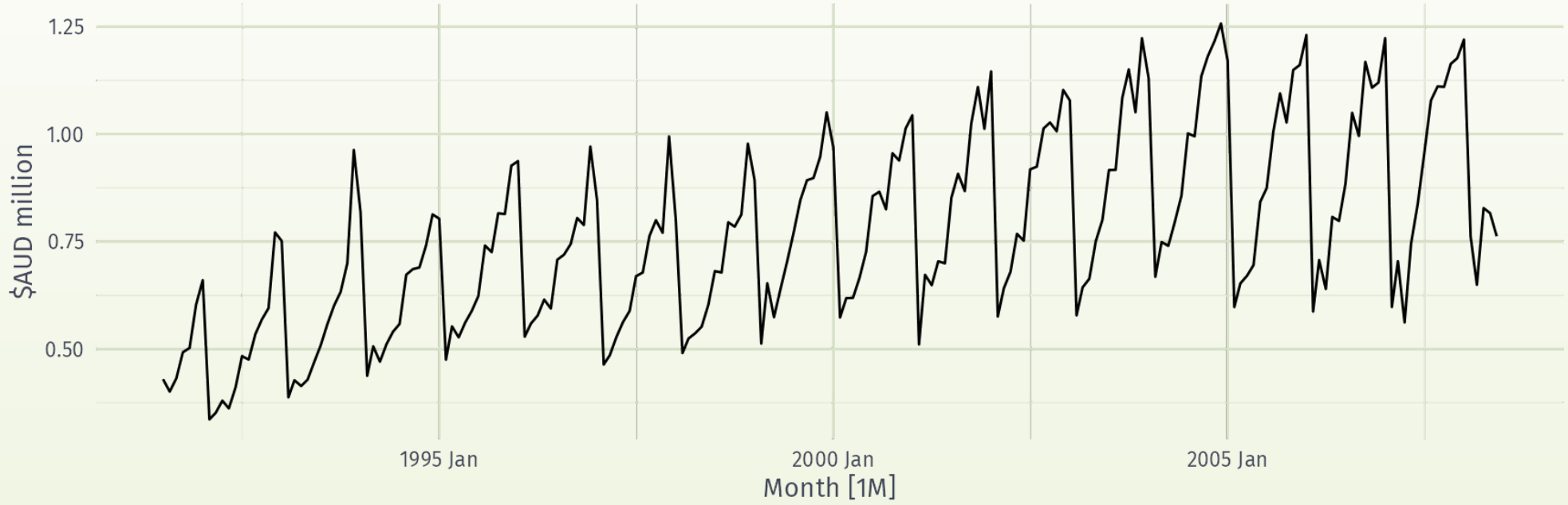
- $m =$ seasonal period (e.g. 12 for monthly data, 4 for quarterly data).
- d first differences, D seasonal differences
- p AR order, q MA order
- P seasonal AR order, Q seasonal MA order

Seasonal and non-seasonal terms combine multiplicatively

Corticosteroid drug sales

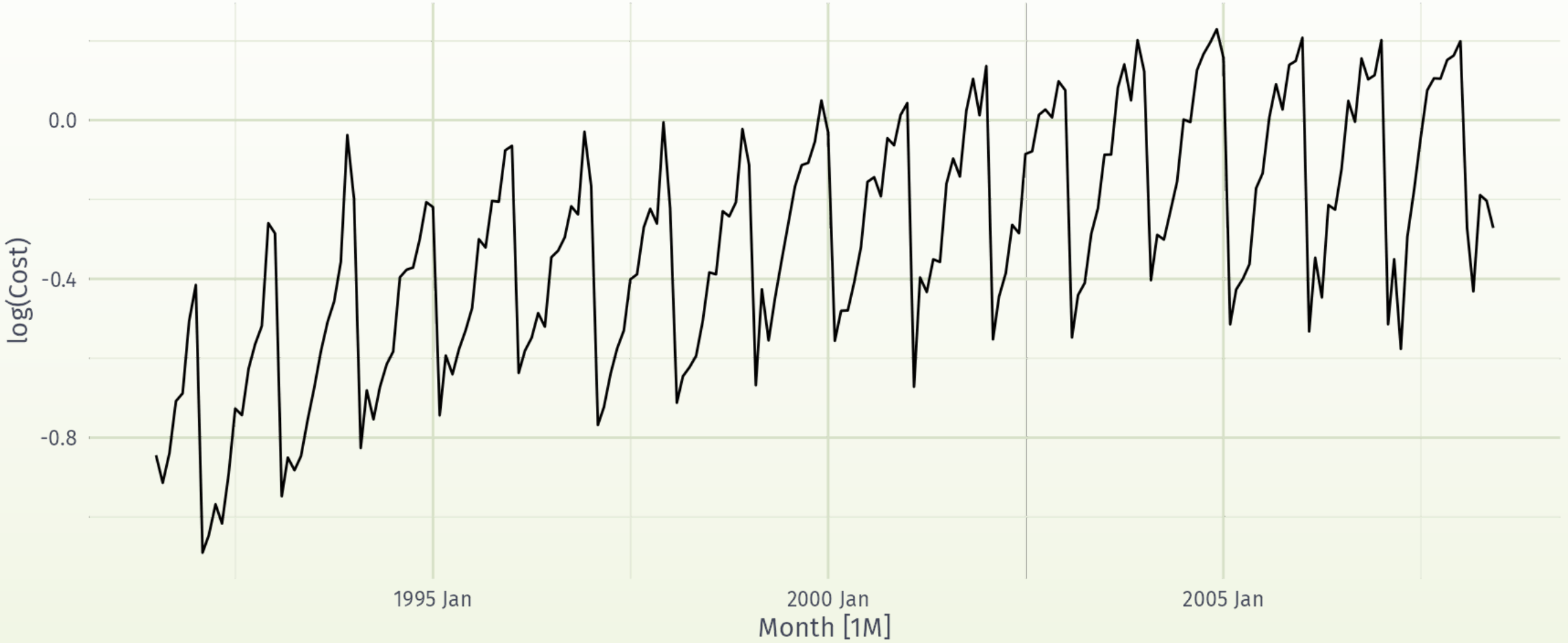
Monthly Australian Pharmaceutical Benefits Scheme (PBS) expenditure on corticosteroid (H02) drugs.

```
h02 <- PBS |>
  filter(ATC2 == "H02") |>
  summarise(Cost = sum(Cost) / 1e6)
h02 |> autoplot(Cost) + labs(y = "$AUD million")
```



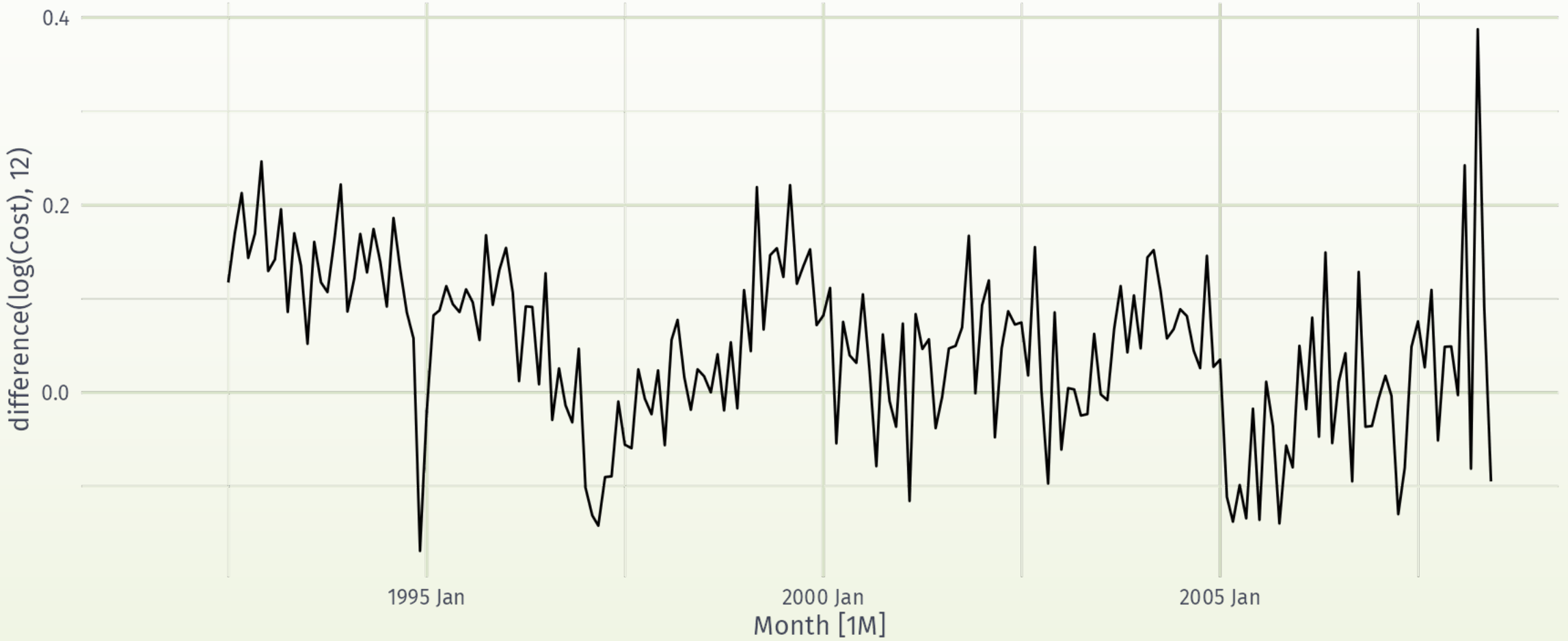
Corticosteroid drug sales: log transform

```
h02 |> autoplot(log(Cost))
```



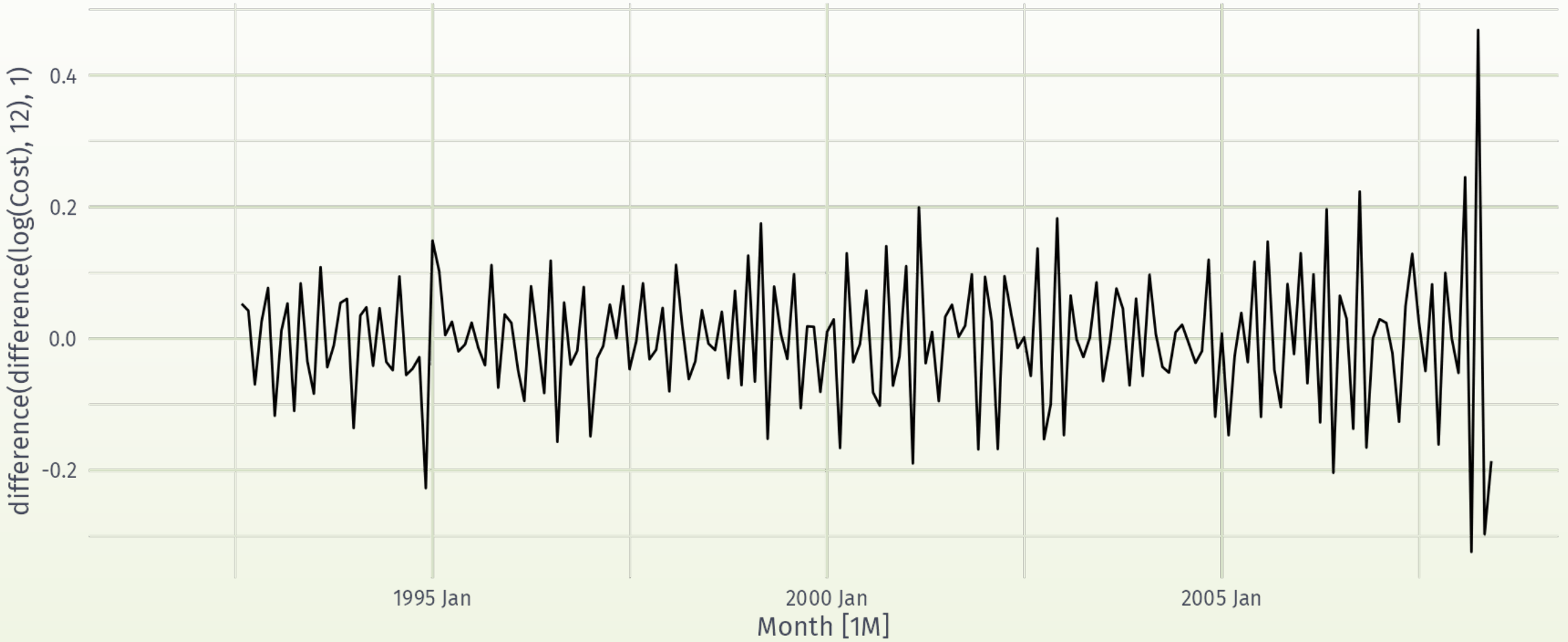
Corticosteroid drug sales: seasonal difference

```
h02 |> autoplot(log(Cost) |> difference(12))
```



Corticosteroid drug sales: seasonal and first difference

```
h02 |> autoplot(log(Cost) |> difference(12) |> difference(1))
```



Corticosteroid drug sales: automatic ARIMA

```
fit_h02 <- h02 |>
  model(auto = ARIMA(log(Cost)))
fit_h02 |> report()
```

```
Series: Cost
Model: ARIMA(2,1,0)(0,1,1)[12]
Transformation: log(Cost)

Coefficients:
          ar1      ar2     sma1
      -0.8491  -0.4207  -0.6401
s.e.   0.0712   0.0714   0.0694

sigma^2 estimated as 0.004387:  log likelihood=245
AIC=-483   AICc=-483   BIC=-470
```

$d = 1$ and $D = 1$ match the differencing above, leaving an AR(2) and a seasonal MA(1) at lag 12.

Corticosteroid drug sales: manual ARIMA

Specify the orders with `pdq()` and `PDQ()`, and the constant with `0 +` (exclude) or `1 +` (include).

```
h02 |>
  model(manual = ARIMA(log(Cost) ~ 0 + pdq(2, 1, 0) + PDQ(0, 1, 1))) |>
  report()
```

```
Series: Cost
Model: ARIMA(2,1,0)(0,1,1)[12]
Transformation: log(Cost)

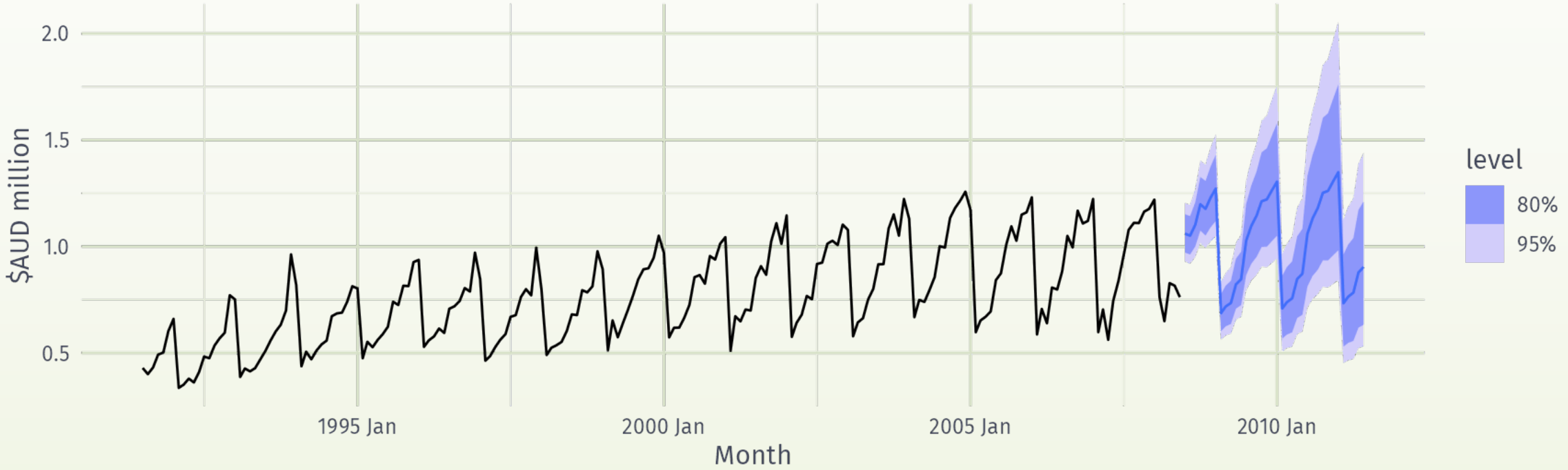
Coefficients:
          ar1      ar2     sma1
      -0.8491  -0.4207  -0.6401
s.e.   0.0712   0.0714   0.0694

sigma^2 estimated as 0.004387:  log likelihood=245
AIC=-483   AICc=-483   BIC=-470
```

Orders can be chosen manually by inspecting the ACF and PACF of the differenced data.

Corticosteroid drug sales: forecasts

```
fit_h02 |>
  forecast(h = "3 years") |>
  autoplot(h02) +
  labs(y = "$AUD million")
```



Corticosteroid drug sales: searching harder

By default `ARIMA()` takes shortcuts to keep model selection fast.

Search options

- `stepwise = FALSE`: fit every model within the order limit, not just the stepwise path.
- `approximation = FALSE`: estimate with the exact likelihood rather than an approximation.
- `order_constraint`: limit the total number of AR and MA terms considered.

A wider search is slower, but may find a model with a lower AICc.

Corticosteroid drug sales: searching harder

```
fit_h02_best <- h02 |>
  model(best = ARIMA(log(Cost),
    stepwise = FALSE,
    approximation = FALSE,
    order_constraint = p + q + P + Q <= 9
  ))
fit_h02_best |> report()
```

```
Series: Cost
Model: ARIMA(4,1,1)(2,1,2)[12]
Transformation: log(Cost)

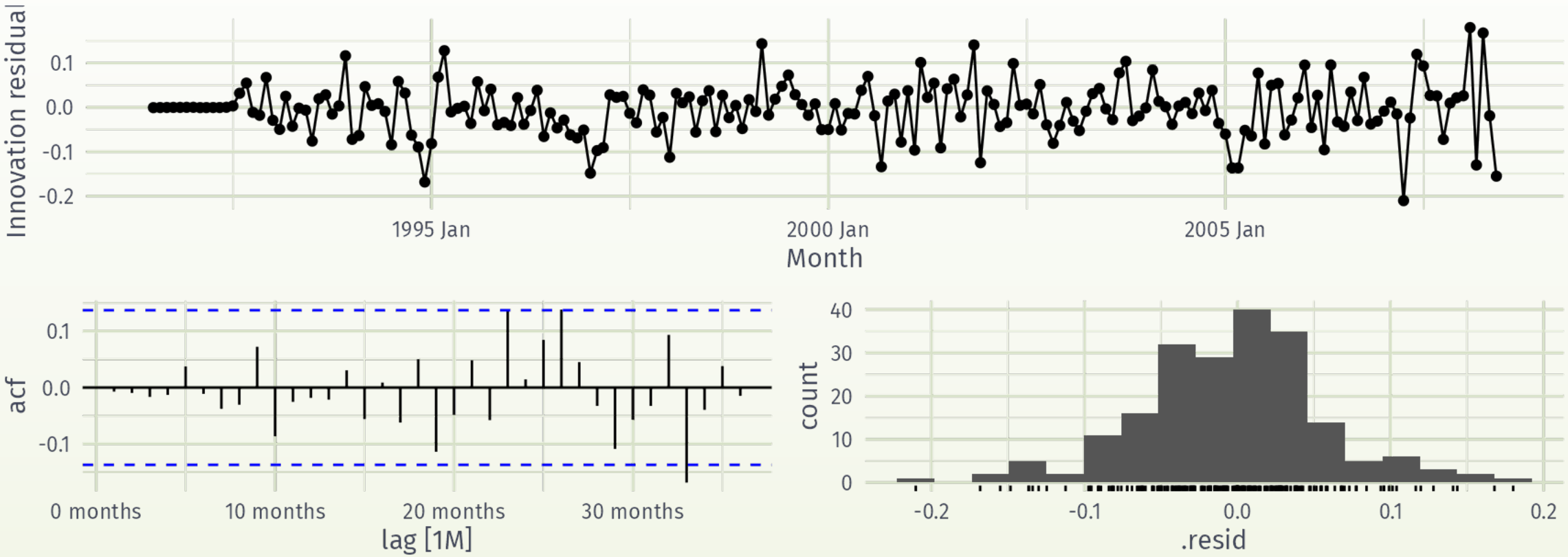
Coefficients:
          ar1    ar2    ar3     ar4     ma1   sar1    sar2    sma1   sma2
      -0.0425  0.210  0.202  -0.227  -0.742  0.621  -0.383  -1.202  0.496
s.e.   0.2167  0.181  0.114   0.081   0.207  0.242   0.118   0.249  0.213

sigma^2 estimated as 0.004049:  log likelihood=254
AIC=-489   AICc=-487   BIC=-456
```

Same differencing and transformation, so the AICc is comparable: -487 vs -483 for the automatic model, using nine coefficients instead of three.

Corticosteroid drug sales: residuals

```
gg_tsresiduals(fit_h02_best, lag_max = 36)
```



Corticosteroid drug sales: residuals

Ljung-Box tests, with `dof` set to the number of estimated coefficients.

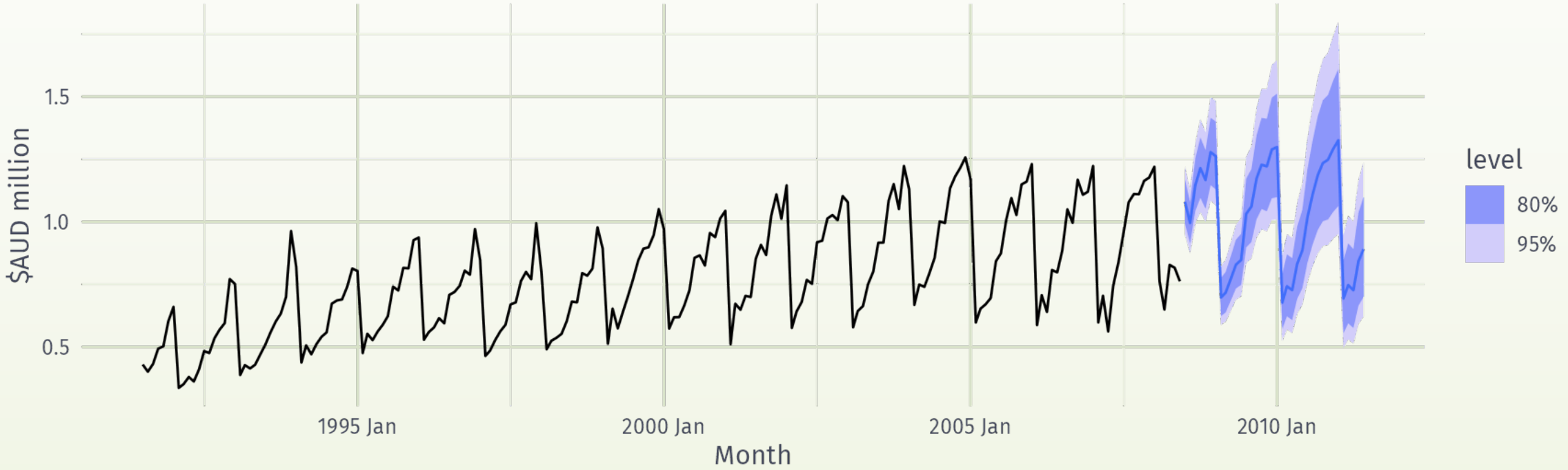
```
bind_rows(
  augment(fit_h02) |> features(.innov, ljung_box, lag = 36, dof = 3),
  augment(fit_h02_best) |> features(.innov, ljung_box, lag = 36, dof = 9)
)
```

```
# A tibble: 2 × 3
  .model lb_stat lb_pvalue
  <chr>    <dbl>     <dbl>
1 auto      59.3   0.00332
2 best      36.5   0.106
```

The automatic model leaves autocorrelation in its residuals; the wider search does not, so we prefer it.

Corticosteroid drug sales: forecasts

```
fit_h02_best |>
  forecast(h = "3 years") |>
  autoplot(h02) +
  labs(y = "$AUD million")
```



SECTION 3 OF 3

Forecast ensembles

1 ARIMA models 2 Seasonal ARIMA models 3 **Forecast ensembles**



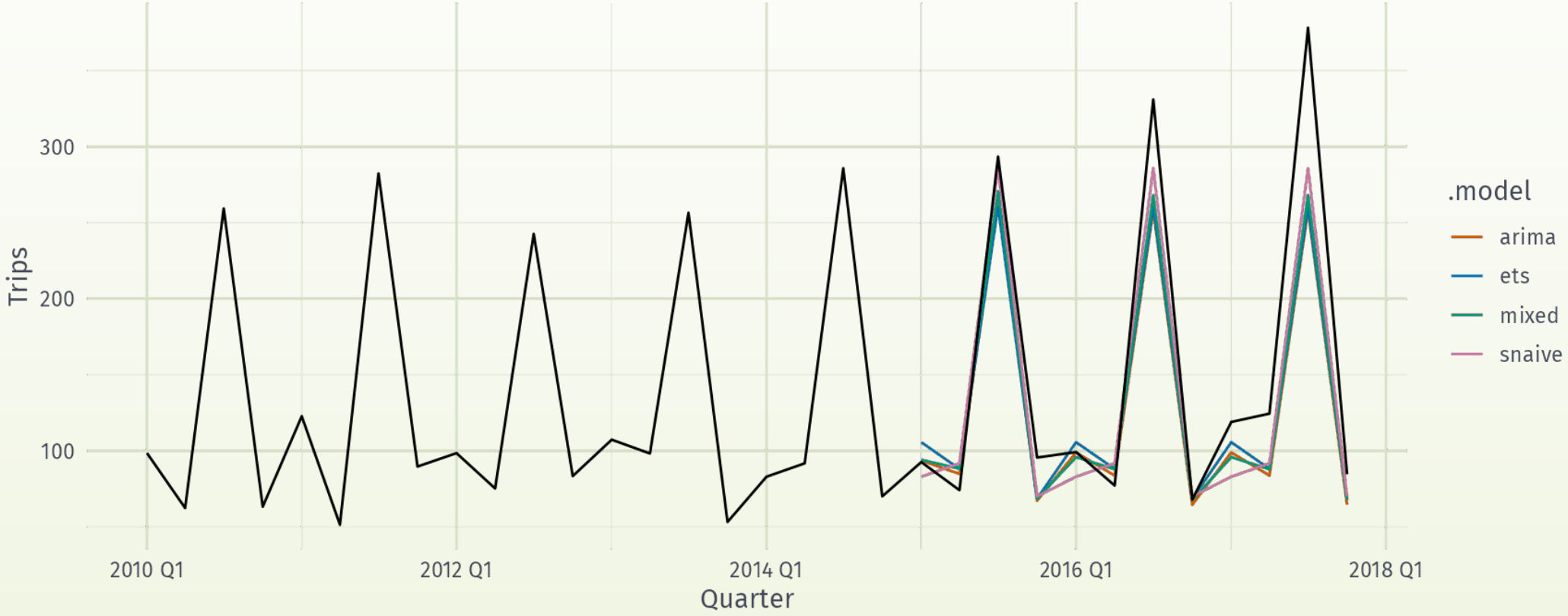
Forecast ensembles

```
train <- tourism |>
  filter(year(Quarter) <= 2014)
fit_tourism <- train |>
  model(
    ets = ETS(Trips),
    arima = ARIMA(Trips),
    snaive = SNAIVE(Trips)
  ) |>
  mutate(mixed = (ets + arima + snaive) / 3)
```

- Fits three models to each of the 304 tourism series (this takes a while).
- Ensemble forecast `mixed` is a simple average of the three fitted models.
- `forecast()` will produce distributional forecasts taking into account the correlations between the forecast errors of the component models.

Forecast ensembles

```
fc <- fit_tourism |> forecast(h = "3 years")
fc |>
  filter(Region == "Snowy Mountains", Purpose == "Holiday") |>
  autoplot(tourism |> filter(year(Quarter) >= 2010), level = NULL)
```



Forecast ensembles

Accuracy on the 2015–2017 test set, averaged across all series.

```
fc |>
  accuracy(tourism) |>
  group_by(.model) |>
  summarise(RMSE = mean(RMSE), MASE = mean(MASE)) |>
  arrange(RMSE)
```

```
# A tibble: 4 × 3
  .model  RMSE  MASE
  <chr>  <dbl> <dbl>
1 mixed   19.8 0.997
2 ets     20.2 1.00
3 snaive  21.5 1.17
4 arima   21.9 1.06
```

Averaging lets errors from different models partially cancel, and reduces the risk of picking the wrong model.